

Czech Technical University in Prague
FJFI - Department of Physics
02ELMA - Electricity and Magnetism
Final exam

30.05.2025 / 14:00pm - 16:00pm

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 120 minutes.
2. The maximum score for the test is 4 points and each question is worth 1 point.
3. Use of any class materials and calculators are not allowed.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. Write your answers in the provided space. If you need more space, ask the instructor for additional sheets.

Question 1

A total charge Q initially dispersed at infinity, is to be brought and uniformly distributed over the surface of a conducting sphere of radius R .

- (a) Express the surface charge density σ on the sphere. — *0.2 pts*
- (b) Express the electrostatic potential $V(\vec{r})$ on the surface of the sphere. — *0.2 pts*
- (c) How much work is required to assemble this configuration? Note that electrostatic energy stored in a surface charge distribution $\sigma(\vec{r})$ with a potential $V(\vec{r})$ is given by $U = \frac{1}{2} \int_S \sigma(\vec{r}) V(\vec{r}) da$ where da is the area element and the integral is calculated over the surface S . — *0.6 pts*

Solution:

- (a) The surface charge density σ is defined as the total charge Q divided by the surface area of the sphere, which is $4\pi R^2$:

$$\sigma = \frac{Q}{4\pi R^2}$$

- (b) The electrostatic potential $V(\vec{r})$ on the surface of the sphere is constant and given by the expression for the potential due to a point charge at a distance R from the charge:

$$V_S = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

- (c) The work required to assemble the configuration is the same as the electrostatic energy stored in the surface charge distribution:

$$U = \frac{1}{2} \sigma V_S \int_S da = \frac{1}{2} \sigma V_S (4\pi R^2) = \frac{1}{2} V_S Q = \frac{1}{2} Q \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{R} \right) = \frac{Q^2}{8\pi\epsilon_0 R}$$

$$W = \frac{Q^2}{8\pi\epsilon_0 R}$$

Question 2

Assume that in the laboratory frame S , there is an electric field $\vec{E} = E_0 \hat{y}$ and no magnetic field $\vec{B} = 0$. An inertial observer S' moves relative to S with constant velocity v along the x -axis. All the axis in both frames are parallel and stay parallel.

- Use the field transformation expressions given below for $\vec{E}$ and $\vec{B}$ to find the electric and magnetic fields $\vec{E}'$ and $\vec{B}'$ as measured in the frame S' . — 0.4 pts
- According to your result in (a), what does an observer in S' measure for the electric field $\vec{E}'$ and the magnetic field $\vec{B}'$? Explain it shortly. — 0.4 pts
- What is the force $\vec{F}' = q(\vec{E}' + \vec{u}' \times \vec{B}')$ acting on the particle in the frame S' ? Note

that the velocity $\vec{u}'$ of the particle is given by the vector $\vec{u}' = \begin{pmatrix} u'_x \\ u'_y \\ u'_z \end{pmatrix}$. — 0.2 pts

Given: The fields $\vec{E}$ and $\vec{B}$ in the reference frame S appear as follows in a frame S' moving with constant velocity v along the x -axis relative to S :

$$\vec{E} = \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} \longrightarrow \vec{E}' = \begin{pmatrix} E'_x \\ E'_y \\ E'_z \end{pmatrix} = \begin{pmatrix} E_x \\ \gamma(E_y - vB_z) \\ \gamma(E_z + vB_y) \end{pmatrix}$$

$$\vec{B} = \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \longrightarrow \vec{B}' = \begin{pmatrix} B'_x \\ B'_y \\ B'_z \end{pmatrix} = \begin{pmatrix} B_x \\ \gamma(B_y + \frac{v}{c^2}E_z) \\ \gamma(B_z - \frac{v}{c^2}E_y) \end{pmatrix}$$

where $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$ is the Lorentz factor, and c is the speed of light in vacuum.

Solution:

- Using the field transformation equations, we can find the electric and magnetic fields in the frame S' :

$$\boxed{\vec{E}' = \begin{pmatrix} 0 \\ \gamma E_0 \\ 0 \end{pmatrix}, \quad \vec{B}' = \begin{pmatrix} 0 \\ 0 \\ -\frac{\gamma v E_0}{c^2} \end{pmatrix}}$$

- In the frame S' , the electric field $\vec{E}'$ is directed along the y -axis with magnitude γE_0 , while the magnetic field $\vec{B}'$ is directed along the z -axis with magnitude $B'_z \equiv -\frac{\gamma v E_0}{c^2}$. This means that an observer in S' sees a stronger electric field due to the Lorentz factor γ , and a magnetic field in the z -direction that arises from the motion relative to the electric field in frame S .

- The force acting on a particle with charge q in the frame S' is given by:

$$\vec{F}' = q(\vec{E}' + \vec{u}' \times \vec{B}')$$

Substituting the expressions for $\vec{E}'$ and $\vec{B}'$, we have:

$$\vec{F}' = q \begin{pmatrix} 0 \\ \gamma E_0 \\ 0 \end{pmatrix} + q \vec{u}' \times \begin{pmatrix} 0 \\ 0 \\ B'_z \end{pmatrix}$$

Calculating the cross product $\vec{u}' \times \vec{B}'$:

$$\begin{pmatrix} u'_x \\ u'_y \\ u'_z \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ B'_z \end{pmatrix} = \begin{pmatrix} u'_y B'_z \\ -u'_x B'_z \\ 0 \end{pmatrix}$$

Thus, the force becomes:

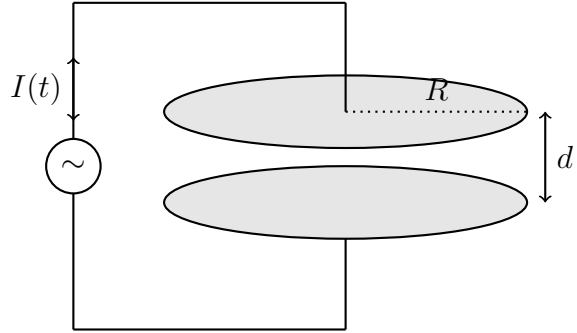
$$\boxed{\vec{F}' = q \begin{pmatrix} u'_y B'_z \\ \gamma E_0 - u'_x B'_z \\ 0 \end{pmatrix}}$$

Question 3

A parallel-plate capacitor with circular plates of radius R and separation d is being charged by a time-varying conduction current

$$I(t) = I_0 \sin(\omega t),$$

through connecting wires, where I_0 is a constant and $\omega \ll 1$ is the angular frequency of the current. The region between the plates is vacuum and $d \ll R$ so that the electric field between the plates is approximately uniform.



- Write down the expression for the electric field $E(t)$ between the plates for as a function of instantaneous charge $Q(t)$ and R . — 0.2 pts
- Calculate the charge $Q(t)$ on the plates by using the integral $Q(t) = \int_0^t I(t') dt'$ and express $E(t)$ in terms of I_0 , R and ω . — 0.3 pts
- Using $\vec{J}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$, find the magnitude of the displacement current density $J_D(t)$ between the plates. What do you get if you consider the multiplication $J_D \pi R^2$? Would you expect it? Explain it briefly. — 0.2 pts
- Use the Ampère-Maxwell law $\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt})$ with a circular integration path of radius r (centered on the axis, lying in a plane midway between the plates) to derive an expression for the magnetic field $B(r, t)$ for $r \leq R$. — 0.3 pts

Solution:

(a) The electric field $E(t)$ between the plates of a parallel-plate capacitor is related to the charge $Q(t)$ on the plates by the formula:

$$E(t) = \frac{\sigma}{\epsilon_0} = \frac{Q(t)}{\epsilon_0 A} = \frac{Q(t)}{\epsilon_0 \pi R^2}$$

(b) The instantaneous charge $Q(t)$ is given by the integral of the current $I(t)$ over time:

$$Q(t) = \int_0^t I(t') dt' = \int_0^t I_0 \sin(\omega t') dt' = \frac{I_0}{\omega} (1 - \cos \omega t)$$

Substituting this expression for $Q(t)$ into the formula for the electric field, we get:

$$E(t) = \frac{I_0}{\epsilon_0 \pi R^2 \omega} [1 - \cos(\omega t)]$$

(c) The displacement current density $J_D(t)$ is related to the time derivative of the electric field by:

$$J_D(t) = \epsilon_0 \frac{\partial E(t)}{\partial t} = \frac{I_0}{\pi R^2} \sin(\omega t)$$

The multiplication $J_D \pi R^2$ gives:

$$J_D \pi R^2 = I_0 \sin(\omega t)$$

This represents the displacement current flowing through the area of the capacitor plates, which is equal to the conduction current $I(t)$ at any instant. It is expected because the displacement current density is defined such that it accounts for the changing electric field in the capacitor, which is necessary to maintain continuity of current in the circuit, especially when the conduction current is not present.

(d) To find the magnetic field $B(r, t)$ for $r \leq R$, let us start with calculating the electric flux for Ampère-Maxwell law:

$$\Phi_E = \int \vec{E} \cdot d\vec{A} = E(t) \cdot \pi r^2$$

Thus, we have:

$$\frac{d\Phi_E}{dt} = \frac{d}{dt} (E(t) \cdot \pi r^2) = \pi r^2 \frac{dE(t)}{dt}$$

Substituting this into the Ampère-Maxwell law, and considering the conduction current inside the capacitor is zero, we get:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(0 + \epsilon_0 \pi r^2 \frac{dE(t)}{dt} \right) = \mu_0 \epsilon_0 \pi r^2 \left(\frac{dE(t)}{dt} \right)$$

The left-hand side is $\oint \vec{B} \cdot d\vec{l} = B(r, t) \cdot 2\pi r$, the expression becomes:

$$B 2\pi r = \mu_0 \epsilon_0 \pi r^2 \left(\frac{I_0}{\epsilon_0 \pi R^2} \sin \omega t \right)$$

After simplifying, and rearranging the terms, we find:

$$B(r, t) = \frac{\mu_0 I_0}{2\pi R^2} r \sin(\omega t) \text{ for } r \leq R$$

Question 4

Consider an electromagnetic plane wave propagating in the $+z$ direction in free space. Its electric field is given by

$$\vec{E}(z, t) = E_0 [\cos(kz - \omega t)\hat{x} + \alpha \sin(kz - \omega t)\hat{y}]$$

where E_0 is a constant, α is a constant concerning the polarization.

- Identify the polarization of the wave for $\alpha = 0$ and $\alpha = 1$; is it linear or circular? — 0.2 pts
 - By using the identity $\vec{B} = \frac{1}{c}\hat{z} \times \vec{E}$, find the magnetic field component $\vec{B}(z, t)$. — 0.2 pts
 - Compute the instantaneous Poynting vector $\vec{S}(z, t) = \frac{1}{\mu_0}\vec{E} \times \vec{B}$. — 0.3 pts
 - Compute the time-averaged Poynting vector $\langle \vec{S} \rangle \equiv \frac{1}{T} \int_0^T \vec{S}(z, t) dt$, where T is the period of the wave. Discuss how the energy flux depends on α . — 0.3 pts
- Given: $\langle \cos^2 \theta \rangle = \langle \sin^2 \theta \rangle = \frac{1}{2}$

Solution:

- For $\alpha = 0$, the electric field is purely in the x -direction, which corresponds to linear polarization along the x -axis. For $\alpha = 1$, the electric field has components in both the x and y directions, which results in circular polarization in the xy -plane, as the two components are equal in magnitude and vary sinusoidally with a phase difference of $\frac{\pi}{2}$.
- Calculating the cross product, we have:

$$\vec{B}(z, t) = \frac{1}{c}\hat{z} \times [E_0 \cos(kz - \omega t)\hat{x} + \alpha E_0 \sin(kz - \omega t)\hat{y}]$$

Using the right-hand rule, we find:

$$\vec{B}(z, t) = \frac{1}{c} [E_0 \cos(kz - \omega t)\hat{y} - \alpha E_0 \sin(kz - \omega t)\hat{x}]$$

- Calculating the cross product, we have:

$$\vec{S}(z, t) = \frac{1}{\mu_0} \left[E_0 (\cos(kz - \omega t)\hat{x} + \alpha \sin(kz - \omega t)\hat{y}) \times \left(\frac{E_0}{c} \cos(kz - \omega t)\hat{y} - \frac{\alpha E_0}{c} \sin(kz - \omega t)\hat{x} \right) \right]$$

Using the properties of the cross product, we find:

$$\vec{S}(z, t) = \frac{E_0^2}{\mu_0 c} (\cos^2(kz - \omega t) + \alpha \sin^2(kz - \omega t)) \hat{z}$$

- The time-average over one period $T = 2\pi/\omega$ gives

$$\langle \cos^2 \theta + \alpha^2 \sin^2 \theta \rangle = \frac{1 + \alpha^2}{2} \implies \langle \vec{S} \rangle = \frac{E_0^2}{\mu_0 c} \frac{1 + \alpha^2}{2} \hat{z}.$$

Hence the average energy flux scales as $(1 + \alpha^2)/2$, giving

- $\alpha = 0$: $\langle S \rangle = E_0^2/(2\mu_0 c)$ (linear),
- $\alpha = 1$: $\langle S \rangle = E_0^2/(\mu_0 c)$ (circular).